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Asset CIA Analysis (10 minutes)

Asset	Confidentiality (1-5)	Integrity (1-5)	Availability (1-5)	Justification <i>Why did you rate it this way?</i>
Pressure/flow readings	3	5	4	If you don't have the accurate data sensor sends (integrity) it can cause catastrophic failure and damage. Availability also matters more than confidentiality because continuously having access to data is important to make sure that sensors function as they should.
Gate control commands	3	4	5	In times of emergency it is important having an access (Availability) to shut off the valve (red button). Also if the dashboard doesn't read the accurate position of the control commands functionality then you are not prepared for an emergency situation or controlling pressure and flow.
Emergency shutoff capability	3	4	5	Again having an access to emergency shutoff is more important than other CIA triad concepts.
Dashboard login credentials	5	4	3	Not leaking login credentials is important for securing systems being accessed by malicious actors.
Consumption/savings data	3	4	2	Having an accurate data results is important for the Hotel and team that controlling the system in order to make sure that sensors functions properly
Leak detection alerts	3	4	5	Knowing that you have access to an alert when it happens is important to stop further damage.

Attack Mapping

For each attack type you learned, identify **one specific scenario** at The Grand Marina:

Attack Type	Specific Scenario	Which Asset Is Targeted	CIA Impact
Eavesdropping	Tap in wired networks in the Grand Marina and eavesdrop sensors packets traveling to broker	Sensors or Management conversation about the whole system	Confidentiality
Spoofing	Add new broker that receives copy of the all the information sensors send and change it	System device such as Broker	Integrity
Replay Attack	Hijack command control session between broker and cloud increase the pressure of the valve	Brooker or Sensor	Integrity
Man-in-the-Middle	Change the messages that are sent to dashboard	Broker	Confidentiality and Integrity
Denial of Service	Flood the Broker with packets	Broker	Availability
Unauthorized Access	Conduct Whaling and steal high ranking personnel credentials and control everything	Employers	Integrity, Availability and Confidentiality

Priority Ranking

Based on your analysis, rank the six attacks from **most dangerous** to **least dangerous** for The Grand Marina:

1. Unauthorized Access - Access to everything can be catastrophic end
2. MITM - Can shut off or increase the pressure to cause colossal damage
- 3.Replay Attack - replay the intended command over and over to break the sensor functionality
- 4.Spoofing - change the information that is read and sent to Dashboard thus deceive everyone that everything is accurate on the sensor reading side

5. Denial of Service - Can't access to sensors when its needed

6. Eavesdropping - Leak confidential information about the Grand Marina to competitor